

		$\frac{3}{4}$ of total area in 1.00s
4	C	Gradient of s-t graph = velocity Between t_1 and t_2 the gradient increases. Hence velocity increases. But the gradient is increasing at a decreasing rate. Hence the acceleration decreases.
5	A	No net external force acting on the system (X, Y and the container), as it is placed on a smooth surface. Hence the centre of gravity of the system does not move.